

Database Schema Documentation

The database schema was designed using PostgreSQL and Prisma ORM to support the maintenance management workflow of the application

Entity-Relationship Diagram (ERD)

The ERD illustrates the relationships between system entities including users, tickets, comments, notifications, and audit logs.

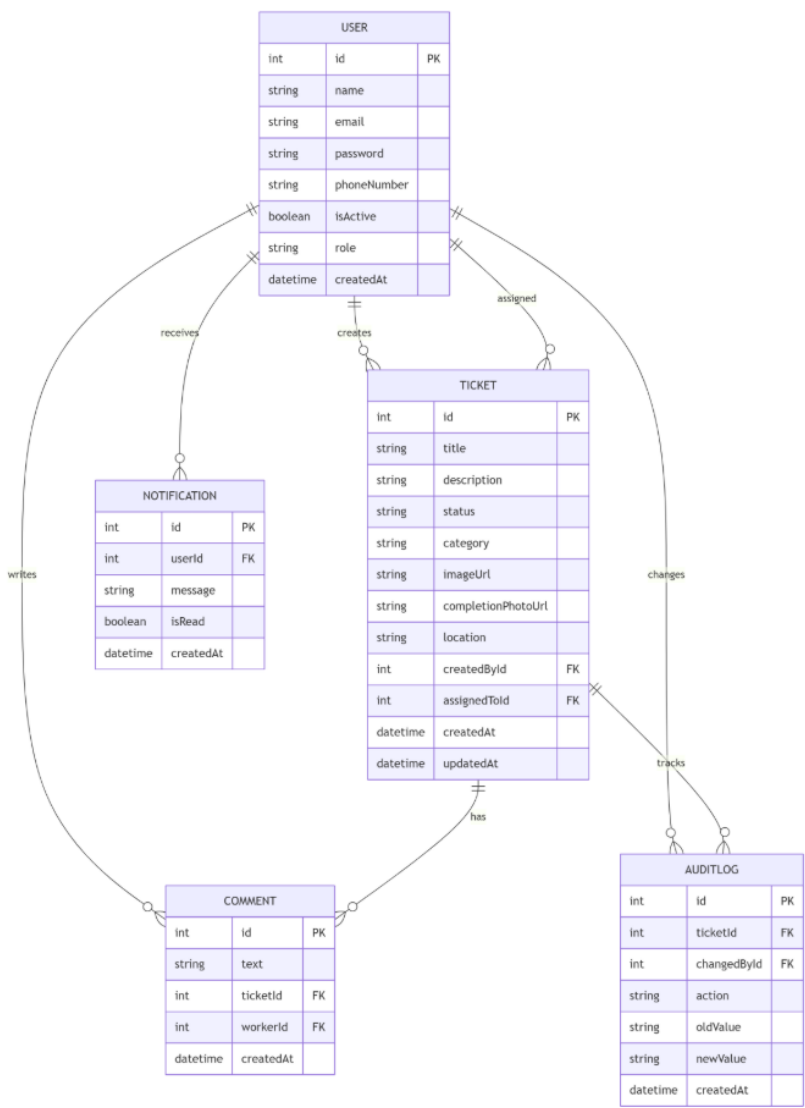


Table definitions

USERS Table

The USERS table stores all authenticated users within the system, including Community Members, Facility Managers, Workers, and Administrators.

Attribute	Type	Description
id	int	Primary key that uniquely identifies each user
name	string	Full name of the user
email	string	Email address used for authentication
password	string	Encrypted password for secure login
phoneNumber	string	User phone number
isActive	boolean	Indicates whether the account is active
role	string	Defines the role of the user within the system
createdAt	datetime	Timestamp of account creation

Relationships

- One user can create multiple tickets.
- One user can be assigned multiple tickets.
- One user can write multiple comments.
- One user can receive multiple notifications.
- One user can create multiple audit log records.

Primary Key (PK): `id` (Integer, Autoincrement).

Unique Constraint: `email` (Must be unique for authentication).

Constraints:

- `name`: Required (String).
- `password`: Required (Encrypted String).
- `isActive`: Boolean, defaults to `true`.
- `role`: Enum (MEMBER, FACILITY_MANAGER, WORKER, ADMIN) .

TICKETS Table

The TICKETS table represents the core entity of the system and stores all submitted maintenance requests.

Attribute	Type	Description
id	int	Primary key that uniquely identifies each ticket
title	string	Short title describing the issue
description	string	Detailed explanation of the maintenance issue
status	string	Current ticket status
category	string	Category of the maintenance issue
imageUrl	string	Uploaded image related to the issue
completionPhotoUrl	string	Uploaded completion image after maintenance
location	string	Reported issue location
createdById	int	Foreign key referencing the reporting user
assignedToId	int	Foreign key referencing the assigned worker
createdAt	datetime	Ticket creation timestamp
updatedAt	datetime	Timestamp of the latest ticket update

Relationships

- Each ticket is created by one user.
- Each ticket may be assigned to one worker.
- One ticket can contain multiple comments.
- One ticket can contain multiple audit log records.

Primary Key (PK): `id` (Integer, Autoincrement).

Foreign Keys (FK):

- `createdById`: References `User.id` (The Community Member who reported the issue).
- `assignedToId`: References `User.id` (The Worker assigned to the task; Nullable until assigned).

Constraints:

- `status`: Enum, defaults to `SUBMITTED`.
- `category`: Enum (`MAINTENANCE`, `CLEANLINESS`, `SUSTAINABILITY`) .
- `imageUrl`: Optional (String).
- `completionPhotoUrl`: Optional (String).

COMMENTS Table

The COMMENTS table stores worker comments and updates related to maintenance tickets.

Attribute	Type	Description
id	int	Primary key that uniquely identifies each comment
text	string	Comment content
ticketId	int	Foreign key referencing the related ticket
workerId	int	Foreign key referencing the worker/user
createdAt	datetime	Comment creation timestamp

Relationships

- Each comment belongs to one ticket.
- Each comment is written by one worker or user.

Primary Key (PK): `id` (Integer, Autoincrement).

Foreign Keys (FK):

- `ticketId`: References `Ticket.id`.
- `workerId`: References `User.id` (The user/worker writing the comment).

Constraints: `text`: Required (String content).

NOTIFICATIONS Table

The NOTIFICATIONS table stores system notifications related to ticket updates and assignments.

Attribute	Type	Description
id	int	Primary key that uniquely identifies each notification
userId	int	Foreign key referencing the related user
message	string	Notification message content
isRead	boolean	Indicates whether the notification has been read
createdAt	datetime	Notification creation timestamp

Relationships

- Each notification belongs to one user.

Primary Key (PK): `id` (Integer, Autoincrement).

Foreign Keys (FK): `userId`: References `User.id`.

Constraints: `isRead`: Boolean, defaults to `false`.

AUDITLOG Table

The AUDITLOG table stores records of important system actions and ticket updates for monitoring and tracking purposes.

Attribute	Type	Description
<code>id</code>	<code>int</code>	Primary key that uniquely identifies each audit log
<code>ticketId</code>	<code>int</code>	Foreign key referencing the related ticket
<code>changedById</code>	<code>int</code>	Foreign key referencing the user who performed the action
<code>action</code>	<code>string</code>	Action performed within the system
<code>oldValue</code>	<code>string</code>	Previous value before modification
<code>newValue</code>	<code>string</code>	Updated value after modification
<code>createdAt</code>	<code>datetime</code>	Timestamp of the logged action

Relationships

- Each audit log belongs to one ticket.
- Each audit log is created by one user.

Primary Key (PK): `id` (Integer, Autoincrement).

Foreign Keys (FK):

- `ticketId`: References `Ticket.id`.
- `changedById`: References `User.id` (The person who performed the action).

Constraints:

- `action`: Required (String describing the system event).
- `oldValue` / `newValue`: Optional (Strings used to track changes)

Enum Definitions:

Role Enum

- MEMBER
- FACILITY_MANAGER
- WORKER
- ADMIN

Status Enum

- SUBMITTED
- ASSIGNED
- IN_PROGRESS
- FINISHED
- FINALIZED
- RESOLVED
- CANCELLED

Category Enum

- MAINTENANCE
- CLEANLINESS
- SUSTAINABILITY

Seed/Sample data